

Proposal for Mach-Zehnder Interferometer Design

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Abstract—This proposal describes the design of Mach-Zehnder Interferometers (MZI) for extracting the group index of a silicon waveguide from the measured free spectral range (FSR). The proposed device uses 500 nm x 220 nm silicon strip waveguides and quasi TE polarization.

I. INTRODUCTION

The objective of this project is to design and fabricate MZIs that can be used to extract group index of silicon waveguides. The group index will be obtained from free spectral range (FSR) and compared with the value obtained from simulations.

II. THEORY

For the MZI, the transfer function is given by:

$$\frac{I_o}{I_i} = \frac{1}{2} [1 + \cos(\beta \Delta L)] . \quad (1)$$

The free spectral range is the spacing between adjacent peaks, $\text{FSR} = \Delta\lambda = \lambda_{m+1} - \lambda_m$. The phase term $\delta = \beta \Delta L$, with $\beta = 2\pi n/\lambda$. Adjacent peaks differ in phase by 2π (high order, $\delta \gg 2\pi$):

$$2\pi = \beta_m \Delta L - \beta_{m+1} \Delta L \Rightarrow \Delta\beta = \beta_m - \beta_{m+1} = \frac{2\pi}{\Delta L} .$$

Approximating β as linear in λ over one period, $\Delta\beta \approx -\frac{d\beta}{d\lambda} \Delta\lambda$, so

$$\Delta\lambda \approx -\frac{2\pi}{\Delta L} \left(\frac{d\beta}{d\lambda} \right)^{-1} .$$

The derivative of β is

$$\frac{d\beta}{d\lambda} = \frac{2\pi}{\lambda} \frac{dn}{d\lambda} - \frac{2\pi n}{\lambda^2} = -\frac{2\pi}{\lambda^2} \left(n - \lambda \frac{dn}{d\lambda} \right) .$$

Inserting this gives

$$\text{FSR} = \frac{\lambda^2}{\Delta L \left(n - \lambda \frac{dn}{d\lambda} \right)} = \frac{\lambda^2}{\Delta L n_g}, \quad n_g = n - \lambda \frac{dn}{d\lambda},$$

so $n_g = \lambda^2 / (\Delta L \cdot \text{FSR})$.

III. MODELING AND SIMULATION

A. Waveguide Design

The waveguide considered for this experiment is a strip waveguide with 500 nm width and 220 nm height, designed for quasi TE polarization. The device will operate around 1550 nm.

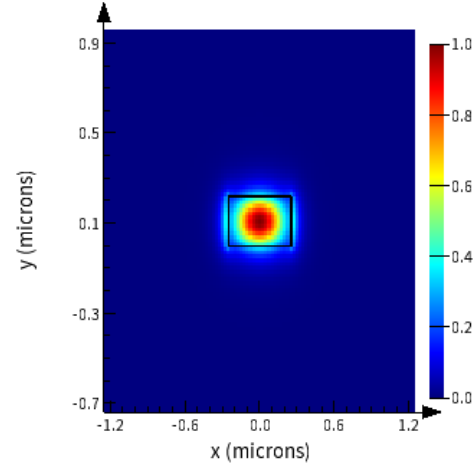


Fig. 1. Simulated fundamental quasi-TE mode of the silicon strip waveguide.

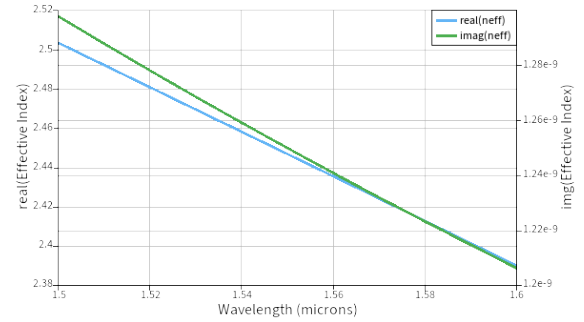


Fig. 2. Simulated effective index of the waveguide.

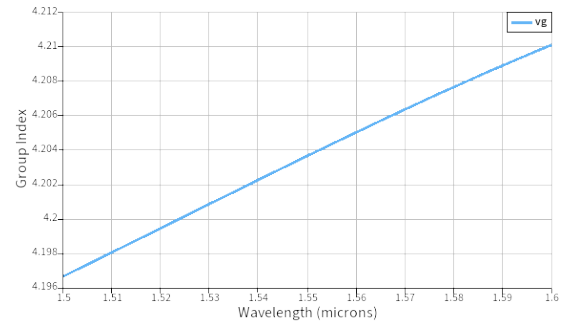


Fig. 3. Simulated group index of the waveguide.

TABLE I
MZI DESIGN VARIATIONS AND EXPECTED FSR

Device	ΔL (μm)	Expected FSR (nm)
MZI1	50	11.356
MZI2	100	5.679
MZI3	150	3.791
MZI4	200	2.8347
MZI5	300	1.898

B. Compact Waveguide Model

The compact waveguide model can be calculated as:

$$n_{eff}(\lambda) = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2. \quad (2)$$

At 1550 nm, the values of $n_1 = 2.44435$, $n_2 = -1.13098$ and $n_3 = -0.04259$.

C. Parameter Variations

Table 1 shows the parameter variations.

D. Transmission Spectrum

Transmission spectrum for $\Delta L = 300$ nm:

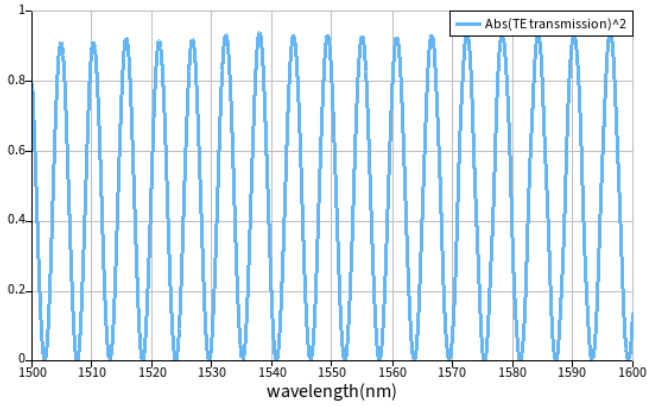


Fig. 4. Transmission spectrum for $\Delta L = 100$ nm